SMA Monitoring API

Sales presentation

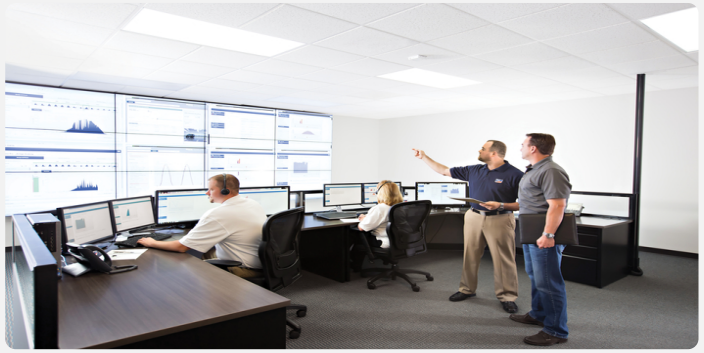
October 1, 2020

Introduction
SMA Monitoring API supports multiple partner use cases

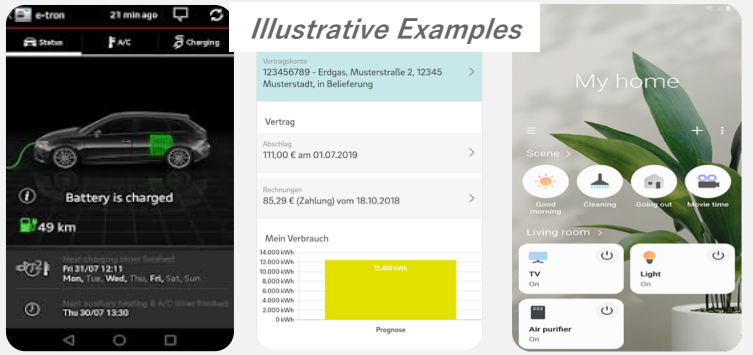
Business Enabler

- SMA Monitoring API is a **B2B business enabler** providing easy to integrate & direct access to SMA PV-system/ device data from SMA portal backends without the need to invest in additional equipment
- SMA Monitoring API is a cloud-based, RESTful, constantly enhanced **Software-as-a-Service (SaaS) proposition** which is currently available in release version 1.3

O&M, SCADA system



eMobility, SmartHome, … Apps



#Process Automation *#Multi-vendor strategies* *#Partner-branded End-user Apps & Portals*

#Reliable Operation *#Flexible integration*

#Cost-efficient *#Secure access* *#Market coverage*

SMA Monitoring API

#SMA Sunny Portal #SMA Sunny Portal powered by ennexOS

One-pager
Flexible integration of SMA plant data into third-party apps

SMA Monitoring API value proposition

Usability	**Monitoring API** provides the following resources: • Plants & Devices • ennexOS Subplants & plant groups • Core/master data • Measurement data • Event logs
Usage of Standards	SMA enforces API security based on **global technical standards** with leading edge oAuth2 cyber security protocol
Performance	**Short response times** are crucial for user acceptance. SMA platform accounts for over 500,000 registered PV systems and +1.5 million connected devices.
Reliability	**High availability** is guaranteed by our contract **SLA** with clear KPIs. If we don't deliver, the API usage is on us.
Ease of Integration	**One API** to access systems on both SMA SunnyPortal classic **and** SMA ennexOS Portal . Our **SMA Sandbox** will jump-start your application integration
Developer Care	**Specialized SMA support** assists in your on-boarding, operation and evolution of our API
Full GDPR compliance	SMA API security is **user-centric**. SMA system owners controls their PV data and must authorize third-party apps access requests
Constant improvements	**Three API releases per year**, SMA adds data and functions requested by our consumers to continuously add and provide more value to you

Full GDPR compliance
SMA offers 2 flows for best & compliant user-experience



- SMA API access flows have been designed to **mitigate non-compliance risks*** for SMA and our partnering third-party application provider
- Our access control is based on a **user-centric protocol for access delegation** to ensure regulatory compliance and provides authorization flows to support the desire of individual users to control how their data is processed
- SMA system owners will have to provide their **explicit (revocable) user consent** to share data from SMA to third-party applications

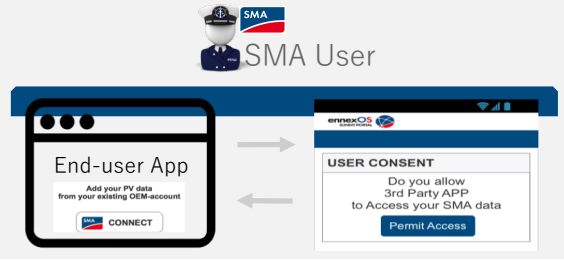
O&M system
(SysOwner is Off-Screen)

O&M agent initiates consent flow for a given SMA system owner from their O&M system



End-user App
(SysOwner is On-Screen)

SysOwner initiates user consent in third-party App e.g., via „Connect SMA“ button



* https://www.csoonline.com/article/3234685/gdpr-fines-how-much-will-non-compliance-cost-you.html

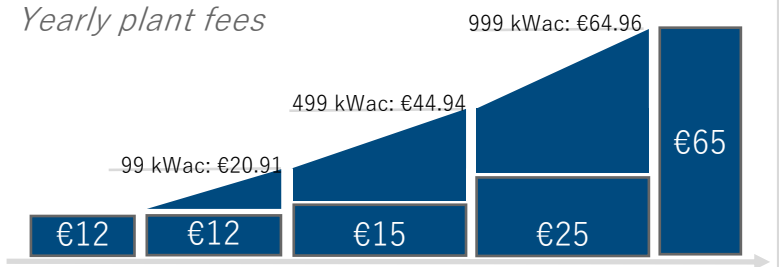
API pricing model
kWac-related monthly base fee starts at €1/plant

Principles

- **No setup fees** for API access / client credential setup
- **Free-of-charge SMA sandbox** for API testing & evaluation
- For productive usage, SMA charges a fair usage, plant size-related **monthly base fee** of €1 to €5.42 / per consumed plant



Monitoring API Pricing
from €1 to €5.42€ per month & plant



SMA Pricing model

Plant size	Base fee	kWac-Factor
1-19	€12	none
20-99	€12	+ €0.09*kWac
100-499	€15	+ €0.06*kWac
500-999	€25	+ €0.04*kWac
>1,000	€65	None

Relative API cost
% on SystemCost / Substitute

SystemSize: 12 kWac
€12/year * 15 years = €180 on lifetime

SystemLife cost- elation:

12 kWp * €1,000/kWp = €12,000
€180 API cost / €12,000: **1.5% over lifetime**

Substitute cost-relation

*Third-party logger: €350
+ On-site installation cost €200
+ Third-party API: setup: €100
+ Third-party API: usage €24/year

€650 one-off cost

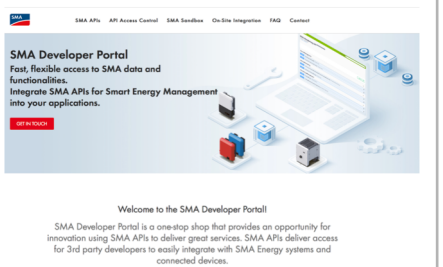
SystemSize: 250 kWac
€30/year * 15 years = €700

SystemLife cost-relation:

250 kWp * €800/kWp = €200,000
€700 API cost / €200,000: **0.035% over lifetime**

Your on-boarding journey
Evaluate and test our API on SMA sandbox environment

Inform



SMA Developer Portal
https://developer.sma.de

including

Swagger



Postman Collection

Postman

Evaluate on SMA Sandbox





TrialForm: 3m SBX access

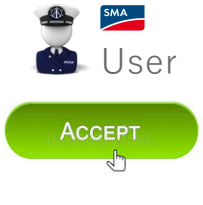
SMA provides SBX credentials

Start coding against SBX

Purchase & Operate









API contract; provide your app data

SMA provides PRD credentials

Request/get SysOwner permit

Use our APIs on production



Thank you!

SMA API access flows
Illustration of end2end user journey

Pre-requisit

SMA Portals

- Installer configured SMA system on SMA Sunny Portal or ennexOS Portal
- All relevant systems have been assigned to a SMA user with „system owner flag“
- On new plants, the system owner has been invited to create SMA user account
- System owner has created a SMA user account

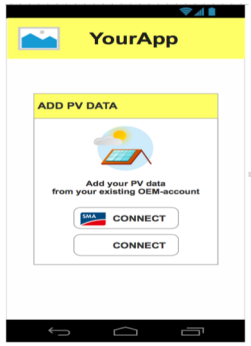
User-consent initiation and decision

End-user App

Sys Owner

O&M System

Agent

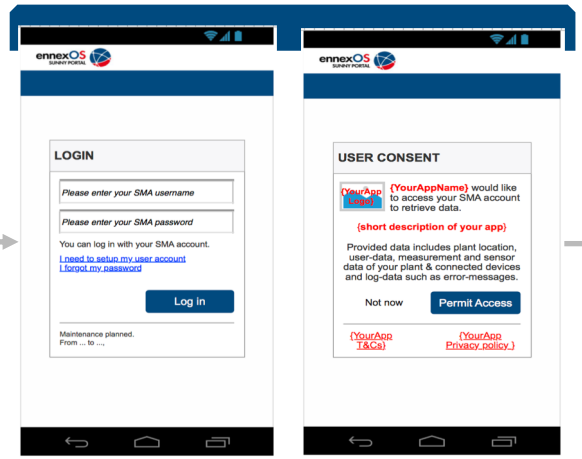




User is redirected to SMA

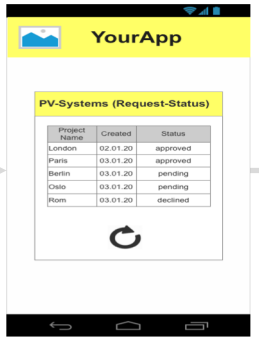
SMA notifies SysOwner

Sys Owner

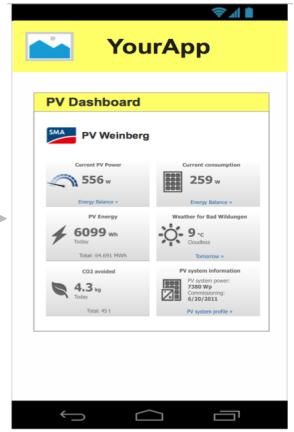


SMA System owner logs-in and provides his/her user-consent for third-party app

End-user app gets token on „accept“



O&M systems polls for user decision



O&M „special“ Asynch

API result